PolarizedTeam at SemEval-2026 Task 9: Detecting Multilingual, Multicultural and Multievent Online Polarization

Nestor Maria[1] Al shrafat Maroan[1] Pește Ioana[1] Daniela Gifu[2,3] Diana Trandabăț[1]

[1]Faculty of Computer Science, "Alexandru Ioan Cuza" University of Iași,
[2]Institute of Computer Science, Romanian Academy- Iasi Branch,
[3]Academy of Romanian Scientists,

{nesmariaes, maroan241, impeste}@gmail.com

daniela.gifu@iit.academiaromana-is.ro

diana.trandabat@info.uaic.ro

Abstract

This paper presents the systems developed for SemEval-2026 Task 9, which targets the detection and categorization of multilingual, multicultural, and multi-event online polarization across 22 languages. To address the challenges posed by linguistic diversity and short, heterogeneous texts, we evaluate several Transformer-based architectures for multilingual polarization detection. Our approach models the task as a multi-label classification problem and incorporates mean pooling for sentence representation, focal loss to mitigate severe label imbalance, and label-wise attention mechanisms to capture polarization-specific linguistic cues. Experimental results show that combining robust multilingual encoders with label-aware modelling substantially improves the detection of polarized content across diverse communities and events[1].

1 Introduction

Online polarization has become a pervasive phenomenon in contemporary digital communication, manifesting through escalating antagonism between social, political, cultural, or identity-based groups. Prior research has shown that polarized discourse often precedes more severe forms of harmful online behavior, including hate speech, harassment, and misinformation, with measurable spillover effects into offline social tensions (Warmsley, 2017). Detecting polarization early is therefore essential for enabling timely interventions, content moderation strategies, and large-scale monitoring of discursive dynamics across communities and events.

The SemEval-2026 Task 9 introduces a multilingual, multicultural, and multievent benchmark designed to capture polarization across 22 languages, reflecting the increasing need for robust cross-lingual models capable of handling heterogeneous, short, and noisy online texts. This challenge aligns with a broader line of research on multilingual social media analysis, including sentiment detection in code-mixed environments (Aparaschivei et al., 2020), multilingual emoji prediction (Alexa et al., 2018), propaganda detection (Ermurachi and Gîfu, 2020), and offensive language identification in low-resource settings (Trandabat et al., 2022). Earlier work on civic identity and discursive markers of social tension (Gîfu and Cioca, 2013), as well as foundational studies on lexical semantics and diachronic variation (Gîfu, 2016), further highlight the complexity of modeling subtle linguistic cues that signal polarization. Together, these studies underscore the need for architectures that can generalize across languages, domains, and cultural contexts.

Despite substantial progress in multilingual NLP, several open challenges remain. Transformer-based encoders such as mBERT and XLM-RoBERTa have demonstrated strong cross-lingual transfer capabilities, yet they struggle with extreme label imbalance, sparse contextual cues in short texts, and the need to capture label-specific linguistic signals, particularly in multi-label settings where different types of polarization may co-occur. Moreover, the diversity of languages included in the POLAR dataset, ranging from high-resource to severely low-resource languages, raises a legitimate question: *To what extent can multilingual encoders, enhanced with task-specific modelling strategies, reliably detect fine-grained polarization across heterogeneous linguistic, cultural, and event-driven contexts?*

This question guides our system design and evaluation.

Our contributions are threefold. (1) We conduct a systematic comparison of multilingual Transformer-based architectures for polarization detection across 22 languages, highlight-

[1]https://github.com/FreaksMind/POLAR-SEMEVAL

ing the strengths and limitations of mBERT, XLM-RoBERTa, and hybrid encoder–decoder configurations. (2) We introduce a modelling pipeline that integrates mean pooling for stable sentence representations, focal loss to address severe label imbalance, and label-wise attention mechanisms that capture polarization-specific linguistic cues in a multi-label setting. (3) We provide an extensive cross-lingual evaluation and error analysis, showing that label-aware modelling substantially improves performance on rare polarization types and low-resource languages.

2 Background

Research on online polarization spans multiple methodological traditions, reflecting the complexity of modeling antagonistic discourse across languages, cultures, and platforms. Early computational approaches relied on supervised classification using lexical, syntactic, and surface-level features, typically employing logistic regression, support vector machines, or random forests (Warmsley, 2017). These studies introduced foundational distinctions between explicit and implicit abuse, as well as targeted versus generalized hostility, but their reliance on shallow features limited their ability to capture nuanced or context-dependent forms of polarization.

As multilingual online communication expanded, representation-learning methods combining multilingual pretraining objectives with explicit alignment losses demonstrated promise for building language-agnostic embeddings suitable for downstream classification and clustering tasks (Mets et al., 2023). Complementary work leveraged pre-trained classifiers to analyse polarization dynamics across platforms and temporal contexts, integrating sentiment, toxicity, stance, and discourse-level indicators to capture broader patterns of online fragmentation (Basile et al., 2019).

The emergence of deep learning and Transformer-based architectures has reshaped multilingual social media analysis. Models such as mBERT and XLM-RoBERTa have become the dominant paradigm for cross-lingual hate speech detection, stance classification, and related tasks, consistently outperforming classical methods, particularly in high-resource languages (Conneau et al., 2020).

Their ability to encode contextual and semantic information across languages has also proven effective in SemEval tasks involving sentiment analysis in code-mixed text (Aparaschivei et al., 2020), multilingual emoji prediction (Alexa et al., 2018), propaganda detection (Ermurachi and Gîfu, 2020), and offensive language identification in low-resource settings (Trandabat et al., 2022). Earlier studies on civic identity and discursive markers of social tension (Ermurachi and Gîfu, 2020), as well as foundational work on lexical semantics and diachronic variation (Gîfu, 2016), further highlight the importance of capturing subtle linguistic cues that often signal polarization.

The POLAR shared task at SemEval-2026 builds on this line of work by introducing a multilingual, multicultural, and multievent benchmark covering 22 languages. The task comprises two subtasks: Subtask 1, a binary classification task determining whether a text contains polarized content, and Subtask 2, a multi-label classification task identifying specific types of polarization (Naseem et al., 2026).

These subtasks present significant challenges, including severe label imbalance, heterogeneous text lengths, sparse contextual cues, and substantial variation across languages.

To address these challenges, we explore multiple multilingual Transformer-based architectures. Our systems include a baseline mBERT model, two XLM-RoBERTa variants (base and large), and a hybrid architecture combining mBERT with an LSTM layer to capture sequential dependencies. Our training strategies are adapted to the characteristics of the POLAR dataset, incorporating techniques for handling class imbalance, label sparsity, pooling strategies, and threshold calibration to improve robustness across languages and polarization types.

Taken together, prior work suggests that effective polarization detection requires models capable of integrating both shared multilingual representations and label-specific linguistic cues. This perspective inspired our approach: rather than relying solely on generic multilingual encoders, we adopt a hybrid modelling strategy that combines robust cross-lingual contextualization with mechanisms explicitly designed to address the structural challenges of the POLAR dataset, namely label imbalance, sparse contextual cues, and the coexistence of multiple polarization types. This conceptual foundation directly shaped the architectures and training strategies presented in the following section.

3 Dataset and Methods

This section introduces the dataset and the methodological approach adopted in our system.

3.1 Dataset

The dataset provided for SemEval-2026 Task 9 consists of short textual snippets collected from online platforms and news or commentary forums across 22 languages: Amharic, Arabic, Bengali, Burmese, Chinese, English, German, Hausa, Hindi, Italian, Khmer, Nepali, Odia, Persian, Polish, Punjabi, Russian, Spanish, Swahili, Telugu, Turkish, and Urdu. Each language is represented by approximately 1,700 to 7,000 instances, with most languages containing around 3-4k samples.

For Subtask 1, each instance includes an ID, the text, and a binary label indicating whether the content is polarized. The overall distribution of polarized versus non-polarized instances is shown in Figure 1, illustrating the imbalance that motivates the use of rebalancing strategies. Text length varies substantially across languages, from very short messages (e.g., Chinese) to longer posts (e.g., Turkish, Khmer, Russian), requiring models to handle heterogeneous input lengths.

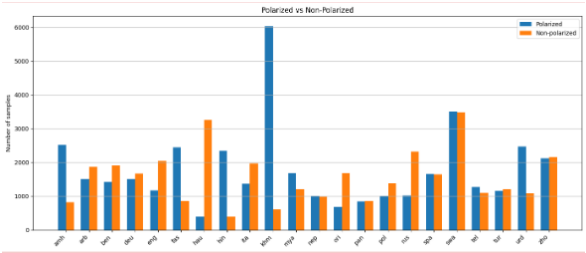

Figure 1: Distribution of polarized vs non-polarized data across all languages

For Subtask 2, each text is annotated with zero or more of five polarization categories: political, racial/ethnic, religious, gender/sexual, and other. The distribution of these categories across languages is presented in Figure 2, highlighting substantial variation and extreme sparsity for several labels. To mitigate this imbalance, non-polarized instances (all-zero labels) are downsampled to 50% of the number of polarized samples. Several languages contain categories with very few or no positive examples, further motivating the use of imbalance-aware training strategies.

Overall, the dataset presents several challenges: multilingual variability, heterogeneous text lengths, severe label imbalance, and sparse positive examples for certain polarization types. These characteristics directly informed our modelling and training decisions.

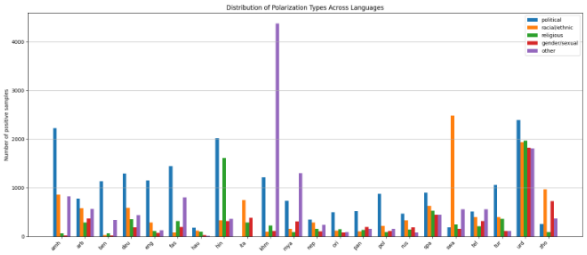

Figure 2: Label distribution across all languages

3.2 Methods

Based on insights from prior work demonstrating the effectiveness of multilingual Transformer-based models, we adopted (Devlin et al., 2019) as a strong and widely used baseline across tasks, and selected XLM-RoBERTa models (Conneau et al., 2020) or their superior multilingual representations and robustness in cross-lingual settings. We describe below the architectures and training procedures adopted for Subtask 1 and Subtask 2 (see Figure 3).

3.2.1 Subtask 1

Subtask 1 is formulated as a binary classification problem, where each text is labeled as polarized or non-polarized. We evaluate three architectures: mBERT, XLM-RoBERTa-Large, and a hybrid mBERT + LSTM model.

Preprocessing and Data Handling

Texts are normalized by removing punctuation, hashtags, and non-informative symbols. Due to the lack of reliable lemmatizers for all 22 languages, no lemmatization is applied. Inputs are truncated or padded to a maximum sequence length of 256 tokens. For each language, the dataset is split into 80% training and 20% validation.

Model Architectures

1. **mBERT baseline**

We fine-tune the multilingual BERT encoder (Devlin et al., 2019), which provides contextualized representations across 100+ languages. However, mBERT is known to struggle with very short texts due to limited contextual information.

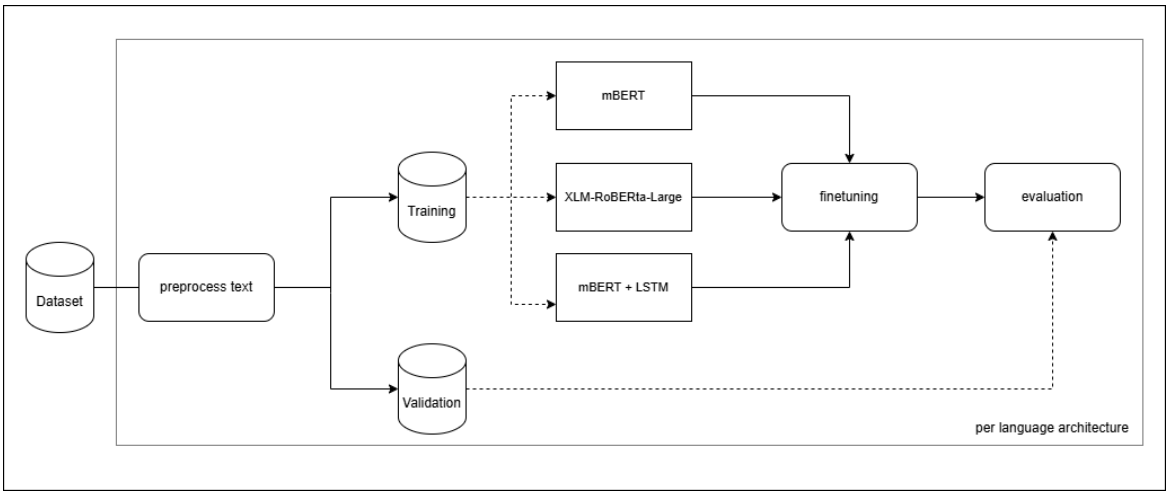

Figure 3: Architecture of the system implemented per language for Subtask 1

2. **XLM-RoBERTa-Large**

This model extends RoBERTa to a multilingual setting and is trained on significantly larger CommonCrawl data. Its increased depth and parameter count allow it to capture richer semantic and cross-lingual patterns (Conneau et al., 2020).

2. **Hybrid mBERT + LSTM**

To complement transformer-based contextual embeddings with sequential modeling, we feed mBERT token embeddings into a unidirectional LSTM layer (Hochreiter and Schmidhuber, 1997) This hybrid architecture aims to capture sequential cues and sentiment shifts that may signal polarization. Adding an attention layer on top of the LSTM did not yield improvements and occasionally led to overfitting, as observed in our validation experiments.

3.2.2 Subtask 2

Subtask 2 requires predicting one or more of five polarization categories for each text: political, racial/ethnic, religious, gender/sexual, and other. The task is substantially more challenging than Subtask 1 due to extreme label imbalance, heterogeneous text lengths, and the presence of languages with very sparse or even missing positive examples for certain categories.

1. **Preprocessing and Label Handling**

Each instance is annotated with a five-dimensional binary vector. To mitigate the dominance of non-polarized examples (all-zero vectors), we downsample these instances to 50% of the number of polarized samples in each language. For languages where a label has zero positive instances, that label is excluded from training to avoid degenerate gradients and unstable loss behavior. All texts are tokenized using the corresponding multilingual tokenizer and truncated or padded to a maximum length of 256 tokens.

2. **Baseline Architecture**

The baseline model uses mBERT (Devlin et al., 2019) as encoder. Token embeddings are aggregated using mean pooling, which provides stable multilingual sentence representations across languages with highly variable text lengths. The pooled embedding is passed through a two-layer feed-forward classifier with ReLU activation and dropout. Training uses BCEWithLogitsLoss with class-specific positive weights computed from the training distribution to counteract label imbalance.

3. **Intermediate Architecture**

To improve sentence-level representations, we replace mBERT with XLM-RoBERTa-Base (Conneau et al., 2020), which offers stronger multilingual contextualization. We concatenate the [CLS] embedding with the mean-pooled embedding to capture both global and token-averaged information. This combined representation is projected through a feed-forward network. To further address imbalance, we adopt Focal Loss, which down-weights easy negatives and focuses learning on rare or difficult positive examples.

4. **Final Architecture: Label-Wise Attention**

To capture label-specific linguistic cues, we introduce a label-wise attention mechanism. For each

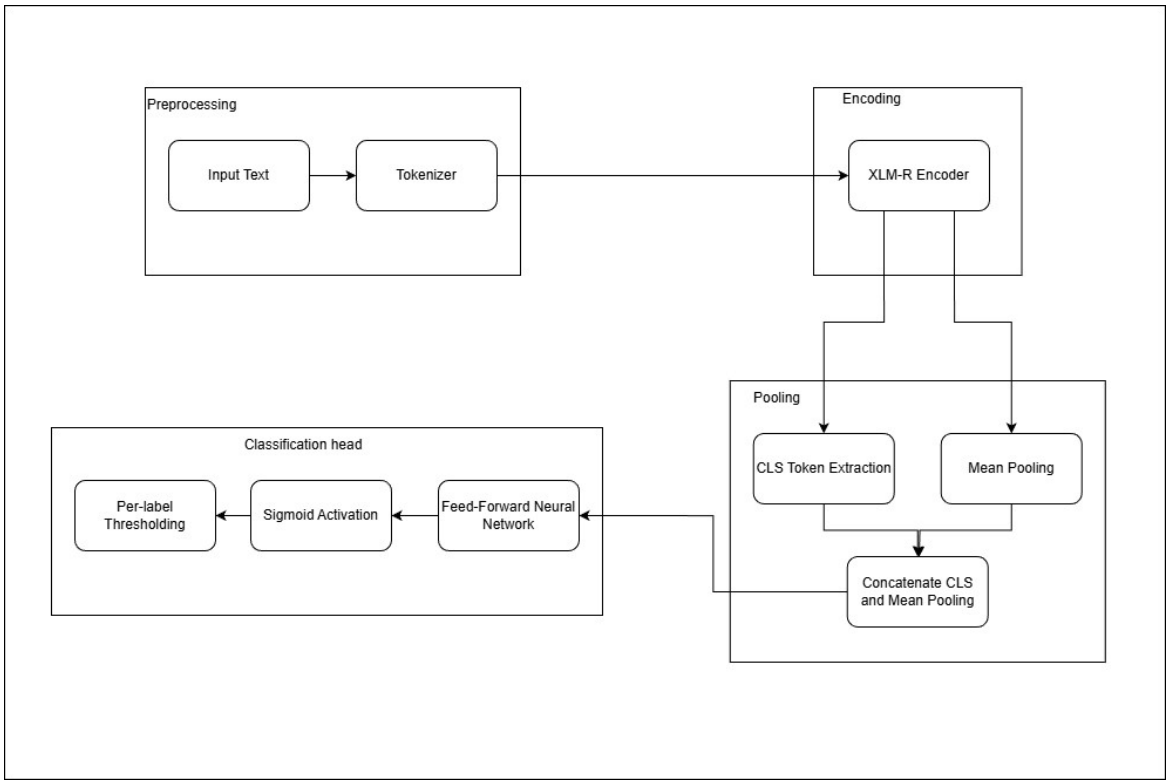

Figure 4: Architecture of the intermediate system used for Subtask 2

polarization category, a dedicated attention head computes token-level importance scores, producing a label-specific sentence embedding. These embeddings are passed through independent linear classifiers, enabling the model to focus on distinct lexical, semantic, and contextual patterns associated with each polarization type. This architecture consistently outperforms the baseline and intermediate models, particularly for rare labels and low-resource languages.

3.3 Experimental Setup

All experiments were implemented in PyTorch using the HuggingFace Transformers library. Models were trained independently for each language to avoid cross-lingual interference and to ensure that performance reflects language-specific distributions. We used the official training, development, and test splits provided by the SemEval-2026 Task 9 organizers.

Training Configuration

We fine-tuned all architectures using the AdamW optimizer with a learning rate of 1×10^{-5} and a linear decay schedule. Gradient clipping with a max-norm of 1.0 was applied to stabilize updates. Batch sizes below 32 produced unstable gradients, while training for more than five epochs consistently led to overfitting; therefore, we used batch sizes of 32–64 and trained for a maximum of five epochs with early stopping based on development loss.

Dropout was applied in all classifier layers to reduce overfitting, and weight decay was set to 0.01. All models used a maximum sequence length of 256 tokens, with truncation applied to longer texts and padding applied to shorter ones. No lemmatization or language-specific preprocessing was performed to maintain comparability across languages.

Loss Functions

For Subtask 1, we optimized binary cross-entropy loss.

For Subtask 2, we used either BCEWithLogitsLoss with class-specific positive weights or Focal Loss, depending on the architecture. Class weights were computed from the training distribution to counteract extreme label imbalance. Focal Loss was implemented following (Lin et al., 2018).

Intermediate System: Mean Pooling and Focal Loss

To enrich sentence-level representations, we em-

ployed XLM-RoBERTa-Base as encoder. Instead of relying solely on the [CLS] token, we computed a mean-pooled representation over all token embeddings, following evidence that mean pooling yields more stable multilingual sentence representations (Del and Fishel, 2022).

Let $H \in \mathbb{R}^{L \times d}$ be the last hidden layer of the transformer, and let $M \in \{0,1\}^L$ denote the attention mask. The pooled representation $h \in \mathbb{R}^d$ is computed as:

$$h = \frac{\sum_{i=1}^{L} H_i * M_i}{\sum_{i=1}^{L} M_i + \epsilon} \tag{1}$$

where L is the sequence length, d is the hidden dimension, and ϵ prevents division by zero.

The final representation is obtained by concatenating the [CLS] embedding with the mean-pooled vector, combining global contextual information with aggregated token-level semantics. This representation is passed through a feed-forward network consisting of a linear layer (768 $\rightarrow$ 256), ReLU activation, dropout, and a final layer producing one logit per label.

Due to extreme label imbalance, we replaced standard cross-entropy with Focal Loss, which focuses training on hard examples and down-weights easy negatives:

For each label $y \in \{0,1\}$ and predicted probability p, the Focal Loss is defined as:

$$\mathcal{L}_{\text{Focal}}(y,p) = (1-p_t)^{\gamma}\, \mathcal{L}_{\text{BCE}}(y,p), \tag{2}$$

where

$$p_t = \begin{cases} p, & \text{if } y = 1, \\ 1-p, & \text{if } y = 0, \end{cases} \tag{3}$$

and γ is a focusing parameter that controls the strength of down-weighting easy examples. This formulation encourages the model to focus on misclassified or uncertain examples, leading to improved decision boundaries for minority classes. The focusing parameter γ controls the degree to which easy examples are down-weighted, encouraging the model to prioritize uncertain or misclassified instances.

This intermediate system achieved an average macro-F1 of 0.495, improving over the baseline but still limited by the use of a shared representation for all labels.

Final System: Label-Wise Attention

To capture label-specific linguistic cues, we extended the architecture with a label-wise attention mechanism and independent classifiers for each polarization type. Each label receives its own attention head, producing a dedicated sentence embedding. These embeddings are passed through dropout and independent linear classification heads. Additionally, lower encoder layers were partially frozen to improve stability in low-resource languages.

This final architecture achieved an average macro-F1 of 0.507, demonstrating clear improvements over both the baseline and intermediate systems.

4 Results

This section presents the performance of the three architectures evaluated for Subtask 1 and Subtask 2 across all 22 languages. We report average accuracy and macro-F1 following the official SemEval evaluation protocol. Results are summarized in Tables 1–6, covering all languages and all model variants.

4.1 Subtask 1 Results

Performance for Subtask 1 is reported in Tables 1, 2 and 3, which present macro-F1 metric for mBERT, XLM-RoBERTa-Large, and the hybrid mBERT+LSTM model across all languages.

Across the board, the hybrid model achieves the strongest results, outperforming both mBERT and XLM-RoBERTa-Large in most languages. This trend is visible in both Table 1 (first set of languages) and Table 2 (remaining languages), where the hybrid architecture consistently yields the highest macro-F1 values.

The mBERT baseline performs well in high-resource languages such as English, Spanish, and Persian, but struggles in languages with short or noisy texts (e.g., Khmer, Burmese), confirming its sensitivity to limited contextual cues.

XLM-RoBERTa-Large improves over mBERT in several mid-resource languages (e.g., German, Hindi, Bengali), but its performance remains inconsistent across the full multilingual spectrum, as shown in both tables.

Overall, Subtask 1 results indicate that combining contextual embeddings with sequential modeling provides the most robust cross-lingual performance.

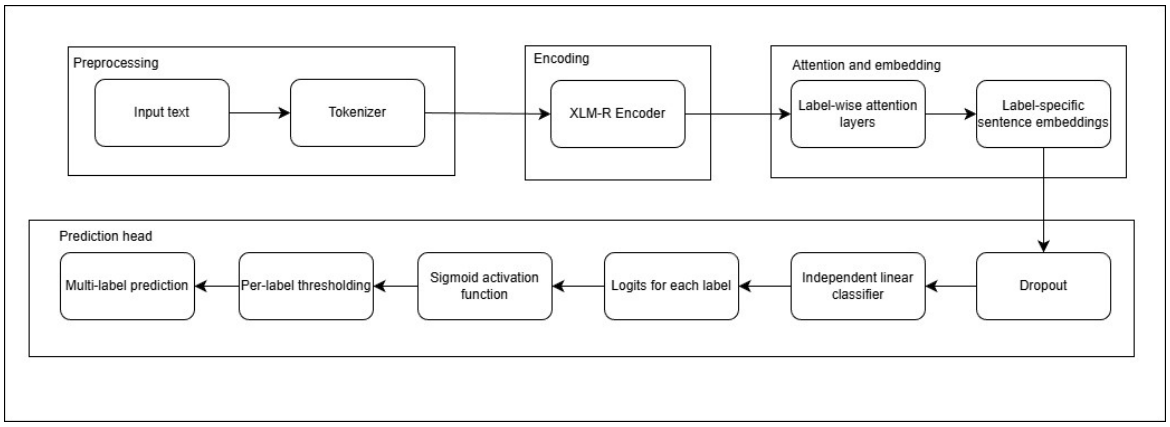

Figure 5: Final System Architecture used for Subtask 2

Model	amh	arb	ben	deu	eng	fas	hau	hin	ita	khm	mya
mBERT	0.423	0.656	0.805	0.636	0.734	0.798	0.794	0.610	0.585	0.189	0.500
XLM-R	0.685	0.757	0.764	0.676	0.762	0.684	0.518	0.733	0.479	0.475	0.660
Hybrid	0.727	0.770	0.840	0.671	0.762	0.837	0.811	0.784	0.648	0.585	0.858

Table 1: F1 scores per model and language (amh → mya)

4.2 Subtask 2 Results

Results for Subtask 2 are also presented in Tables 4, 5 and 6, covering all 22 languages. The task is substantially more challenging due to extreme label imbalance and the multi-label nature of the problem.

The baseline mBERT model achieves an average macro-F1 of 0.453, with strong performance in high-resource languages but significantly lower scores in languages with sparse positive labels.

The intermediate XLM-RoBERTa-Base model, which integrates both [CLS] and mean-pooled embeddings, improves the average macro-F1 to 0.495. This improvement is visible across both tables, particularly in Bengali, Hindi, German, and Spanish.

The final label-wise attention architecture achieves the best overall performance, with an average macro-F1 of 0.507. This model shows consistent gains across languages with highly skewed label distributions, such as Amharic, Hausa, and Telugu.

4.3 Cross-Lingual Trends

A detailed inspection of Tables 1–6 reveals several cross-lingual patterns:

1. High-resource languages (English, Spanish, Persian, Urdu) show stable performance across all models, with incremental gains from the attention-based architecture.

2. Low-resource languages (Hausa, Amharic, Nepali) benefit disproportionately from the hybrid and attention-based models, confirming the importance of label-aware mechanisms.

3. Languages with short texts (Chinese, Khmer, Burmese) remain challenging for all models, though the final architecture shows the most resilience.

4. Morphologically rich languages (Turkish, Russian, Polish) exhibit moderate improvements with XLM-RoBERTa and the final architecture, suggesting that deeper multilingual encoders help capture complex morphological patterns.

4.4 Model Comparison and Architecture Insights

Figure 5 presents the architecture of the final system used for Subtask 2. This visualization clarifies why the label-wise attention model outperforms the baseline and intermediate systems: each label receives a dedicated attention head and classifier, enabling the model to capture fine-grained linguistic cues that generic pooling cannot.

Table 6 summarizes the aggregated results across all languages and models. These aggregated metrics confirm the superiority of the final architecture, which consistently achieves the highest macro-F1

Model	nep	ori	pan	pol	rus	spa	swa	tel	tur	urd	zho
mBERT	0.678	0.458	0.682	0.577	0.515	0.640	0.799	0.661	0.622	0.571	0.774
XLM-R	0.799	0.723	0.653	0.722	0.604	0.592	0.639	0.703	0.773	0.650	0.845
Hybrid	0.880	0.752	0.740	0.729	0.741	0.696	0.799	0.872	0.713	0.760	0.897

Table 2: F1 scores per model and language (nep → zho).

Model	Average Accuracy	Average F1 Score
mBERT	0.6876	0.6235
XLM-RoBERTa-Large	0.7198	0.6775
mBERT + LSTM	0.8069	0.7673

Table 3: Average accuracy and F1 scores across all languages for different models.

across both subtasks. Across all six tables, the following conclusions emerge: Hybrid architectures outperform pure Transformer models in binary polarization detection (Subtask 1); Label-wise attention provides the largest gains in multi-label classification (Subtask 2); Focal Loss significantly improves performance on rare polarization categories; Cross-lingual robustness increases with architectures that explicitly model label-specific cues.

Performance patterns are consistent across all 22 languages, as documented in Tables 1-6.

These findings validate our modelling choices and demonstrate the effectiveness of combining multilingual contextualization with task-specific mechanisms for fine-grained polarization detection.

5 Discussions

Although our systems achieve consistent improvements across both subtasks, several limitations remain. The most persistent challenge is extreme label imbalance, which continues to affect rare categories despite the use of Focal Loss and label-wise attention. Performance also drops sharply in languages with very short or noisy texts, where contextual cues are insufficient for Transformer-based encoders. In addition, training models independently per language prevents cross-lingual transfer that could benefit low-resource settings. Finally, attention heads in the final architecture show higher variance across languages, suggesting sensitivity to sparse training signals. These limitations indicate that multilingual polarization detection remains difficult, particularly for low-resource languages, rare labels, and short-text scenarios.

6 Conclusion

This work presented a multilingual system for detecting online polarization across 22 languages, addressing both binary detection (Subtask 1) and fine-grained multi-label classification (Subtask 2). Across both subtasks, our experiments show that combining strong multilingual encoders with task-specific modelling strategies leads to consistent improvements over baseline Transformer architectures.

For Subtask 1, the hybrid mBERT+LSTM model proved most effective, demonstrating that sequential modelling complements contextual embeddings when dealing with short, heterogeneous texts. For Subtask 2, the integration of mean pooling, imbalance-aware optimization, and label-wise attention significantly improved the detection of rare polarization categories, yielding the highest macro-F1 across languages.

Despite these gains, challenges remain, particularly in low-resource languages, highly imbalanced label distributions, and languages with extremely short or noisy texts. These limitations highlight the need for future work on cross-lingual transfer, data augmentation, and architectures tailored to sparse or low-context inputs.

Overall, our results demonstrate that multilingual polarization detection benefits from architectures that combine robust cross-lingual representations with mechanisms explicitly designed to capture label-specific linguistic cues. The proposed systems provide a solid foundation for advancing polarization analysis in diverse linguistic and cultural settings.

Acknowledgments

This work was carried out partially within the project "Tools for Processing Online Texts Specific to Cultural and Scientific Diplomacy", funded by the Academy of Romanian Scientists.

Model	amh	arb	ben	deu	eng	fas	hau	hin	ita	khm	mya
mBERT	0.333	0.441	0.222	0.491	0.392	0.578	0.100	0.776	0.341	0.258	0.527
XLM-R (CLS+mean pooling and Focal Loss)	0.488	0.531	0.209	0.479	0.337	0.621	0.077	0.804	0.343	0.652	0.500
XLM-R (label-wise attention)	0.511	0.518	0.287	0.503	0.343	0.592	0.089	0.801	0.361	0.661	0.551

Table 4: F1 scores per model and language (amh → mya).

Model	nep	ori	pan	pol	rus	spa	swa	tel	tur	urd	zho
mBERT	0.742	0.149	0.385	0.379	0.431	0.606	0.470	0.437	0.436	0.758	0.704
XLM-R (CLS+mean pooling and Focal Loss)	0.713	0.481	0.345	0.439	0.419	0.583	0.462	0.468	0.449	0.751	0.734
XLM-R (label-wise attention)	0.728	0.468	0.401	0.417	0.533	0.595	0.453	0.386	0.445	0.750	0.753

Table 5: F1 scores per model and language (nep → zho).

Model	Average F1 Score
mBERT	0.453
XLM-R (CLS+mean pooling and Focal Loss)	0.495
XLM-R (label-wise attention)	0.507

Table 6: Average F1 scores across all languages.

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